Rearranging Equations

LLE – Mathematics and Statistics
ECO-5007A

Skills - Rearranging Equations

In each of the following expressions, make x the subject of the equation.

1. $x + y = 12$
2. $5x - 4 = y$
3. $ax - 3by = 9$
4. $y = mx + c$
5. $2 + 5xy = 6$
6. $9x + 6y = 4x + 1$

Tip: Gather all terms containing x on one side of the equation and all other terms on the opposite side. Then, factor out x.

7. $ax + by = 4x + 1$
8. $ax - by = bx - 8$
9. $\dfrac{2ty + mx}{ty - mx} = 1$
10. $\dfrac{ty + mx}{ty - 3mx} = k$
11. $9x^2 = 4y$
12. $x^3 - 5m = t$

13. $x^2 - 2x = 15$

Tip: Rearrange this into the standard quadratic form $ax^2+bx+c=0$ and then consider using the quadratic formula or factoring.

14. $x^2 = 2 - qx$

15. $x^2 - 8 = kx$

    - For a bonus mark on this part: what must be true for this expression to have real solutions?

Rearranging with Indices

This section uses the index law of $x^a x^b = x^{a+b}$ and that a number multiplied by its reciprocal is 1. This is useful for more difficult indices.

Example: Rearrange to make x the subject of $x^{\frac{4}{9}} = y^{\frac{1}{3}}$
Solution:

$$x^{\frac{4}{9}\times\frac{9}{4}} = y^{\frac{1}{3}\times\frac{9}{4}}$$
$$x = y^{\frac{3}{4}}$$

Practice Problems:

Rearrange the following to make x the subject:

1. $x^{\frac{3}{4}} = y^{\frac{1}{2}}$

2. $x^{\frac{3}{4}} = y^{\frac{1}{2}} - 2$

3. $5x^8 - y = 0$

4. $x^{\frac{3}{4}}y = 8my$

5. $2x^{2.5} = 64$

6. $x^2y^3x^5y^2 = a$

7. $0.5x^{-0.5}y^{-0.5} \times 0.5x^{-0.5}y^{0.5} = 4$

8. $0.8y^{-0.2}x^{0.2} \times 0.2y^{0.8}x^{-0.8} = 10$

9. $\dfrac{\alpha y^{\alpha-1}x^{\beta}}{\beta y^{\alpha}x^{\beta-1}} = k$

Multivariate Optimisation Rearranging Practice

Zy is a pet dog, who gains utility from eating and from playing catch. Her utility function is defined as:

$$U(E, C) = E^{0.6}C^{0.4}$$

Where E is the number of hours eating and C is the number of hours playing catch.

During the day, Zy has a maximum of 8 hours for eating and catching.

For maximum utility, how many hours should Zy spend eating and catching, to the nearest minute?

Part of a solution:

Constraint function:

$$E + C = 8$$

Lagrangian:

$$L = E^{0.6}C^{0.4} - \lambda(8 - ...)$$

1. Replace the dots with the rearranged constraint function.

First Order Conditions:

$$\frac{dL}{dE} = 0.6E^{-0.4}C^{0.4} - \lambda = 0$$

$$\frac{dL}{dC} = 0.4E^{0.6}C^{-0.6} - \lambda = 0$$

2. For both first order conditions, rearrange to make λ the subject of an equation in E and C.

Dividing the Equations:

You then see that they have divided these two equations by one another:

$$\frac{0.6E^{-0.4}C^{0.4}}{0.4E^{0.6}C^{-0.6}} = \frac{\lambda}{\lambda}$$

3. Make C the subject of this equation. You will need to use rules of indices.

Tip: Remember that $\frac{x^a}{x^b} = x^{a-b}$ *and* $\frac{1}{x^n} = x^{-n}$*.*

4. Substitute your equation for C into the constraint function $E + C = 8$.

5. Solve this equation for E.

6. Hence find the values of E and C that maximise utility for Zy and convert your answer to hours and minutes (e.g., 3.5 hours is 3 hours and 30 minutes).

Challenge Problem: Multivariate Optimisation Question

This last question uses more skills than this workshop but is made available for further practice of ECO-5007A work.

A utility function is given:

$$U = x^{\frac{1}{2}}y^{\frac{1}{2}}$$

With constraint:

$$5x + 4y = 20$$

Find values for x and y that maximise the utility function.